

Q.2] Air Traffic Control System – AVL Tree Analysis

1. Problem

- The AVL tree is initially keyed by aircraft transponder code.
- Searching by transponder code is fast, but it does not help find the nearest aircraft.
- Changing the key to aircraft distance makes spatial searching easier.
- However, aircraft distance changes continuously.
- Every distance update requires deletion + reinsertion into the AVL tree.

2. Performance Issue

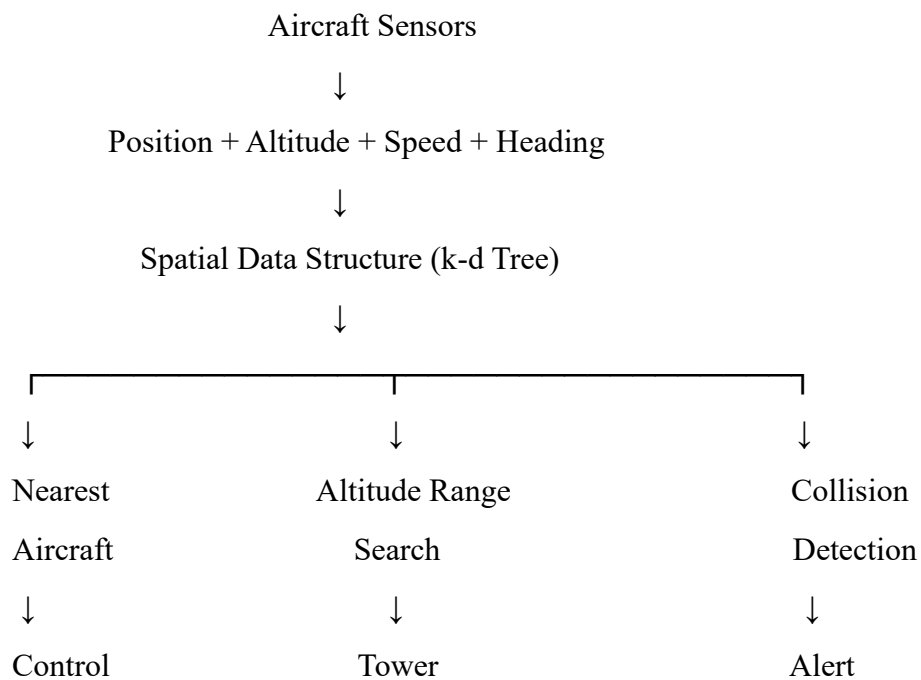
- Number of aircraft = 500
- Each aircraft updates every 2 seconds.
- Each update requires approximately 2 AVL rebalancing operations.
- This produces about $500 \times 2 = 1000$ rebalancing operations per second.
- This consumes around 30% CPU time, making the system inefficient.

3. Better Data Structure

A better solution is to use a spatial data structure, such as a k-d tree (k-dimensional tree).

- Store aircraft using their X and Y coordinates.
- Nearest-aircraft queries can be performed efficiently.
- Spatial regions can be searched without ordering everything by constantly changing distance.
- Altitude can be included as an additional dimension or maintained in a separate index.
- Position updates can be handled more efficiently than repeatedly modifying an AVL tree keyed by distance.

4. Proposed System



5. Conclusion

The AVL tree is efficient for static or slowly changing ordered data, but it is not ideal for aircraft positions that change every 2 seconds. A k-d tree or another spatial indexing structure is more appropriate because the primary operations are nearest-neighbor and spatial-range queries, not searches by transponder code.